



Smart Agriculture: Crop Recommendation System

AI-Driven Yield Optimization using Random Forest Classification

Ashliya M Riyaz

Nivedh P D

Bainty Kaur Chugh

THE AGRICULTURAL CHALLENGE



Soil Inconsistency

Farmers struggle with imbalanced nutrient ratios (N, P, K), leading to poor crop health and reduced soil fertility.



Climate Instability

Unpredictable temperature and rainfall patterns make traditional crop selection methods unreliable.

Economic Risks

Choosing the wrong crop for a specific soil type leads to massive financial losses and wasted labor capital.

PREDICTIVE PARAMETERS

Primary Nutrients: Monitoring Nitrogen, Phosphorus, and Potassium levels.

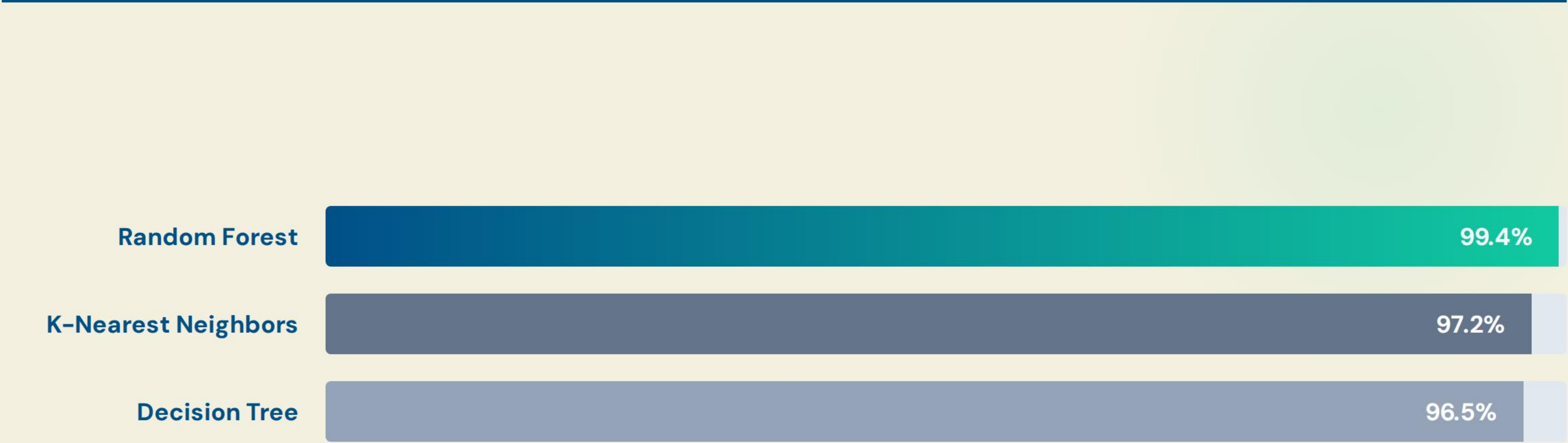
Climatic Metrics: Temperature and Relative Humidity percentages.

Environmental: Soil pH levels and seasonal Rainfall volume (mm).

Dataset Scale: 2,200 Observations | 22
Unique Crop Categories

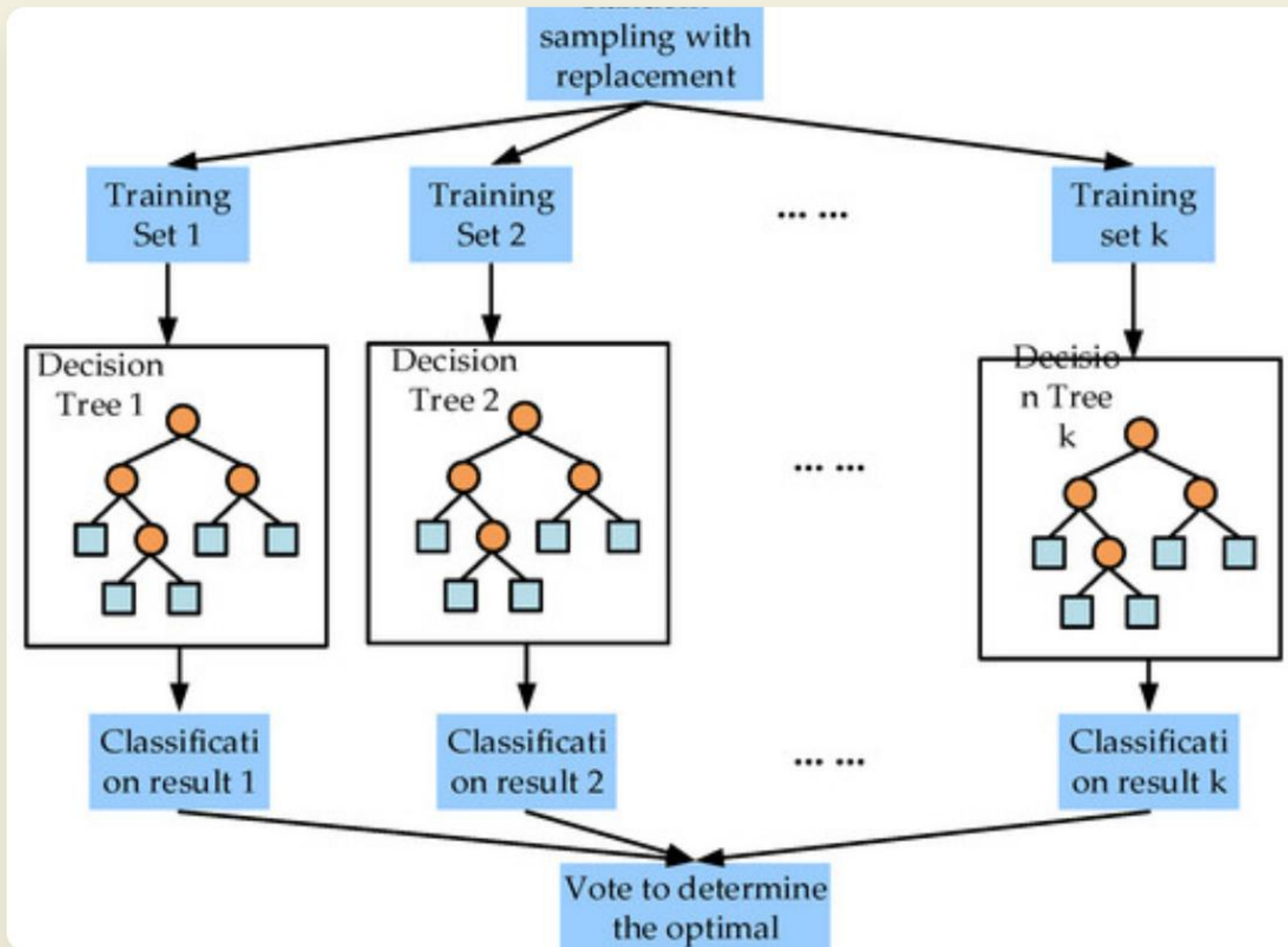


MODEL PERFORMANCE BENCHMARKING



Random Forest was selected as the final model due to its superior accuracy and resistance to dataset noise.

CORE ALGORITHM: RANDOM FOREST



- ✔ **Ensemble Logic:** Uses majority voting from multiple trees to ensure prediction stability.
- ✔ **Non-Linearity:** Efficiently captures complex interactions between pH, humidity, and rainfall.
- ✔ **Reliability:** Standard deviation in accuracy remained extremely low during K-fold validation.

DETERMINANTS: FEATURE IMPORTANCE



- Rainfall (25%)
- Humidity (20%)
- Potassium - K (15%)
- Phosphorus - P (15%)
- Nitrogen - N (10%)
- Others (15%)

DEPLOYMENT ARCHITECTURE

Platform Layer	Technology Used	Implementation Role
Development	Python / Google Colab	Data Engineering & Model Training
Machine Learning	Scikit-Learn	Classifier Implementation & Evaluation
Web Framework	Streamlit	Dynamic Front-end for Soil Inputs
Persistence	Pickle	Bundling Model & Label Encodings
Deployment	Ngrok	Secure Tunneling for Public Access

INTERACTIVE USER INTERFACE

Real-Time Prediction

The Streamlit app allows farmers to input environmental data via interactive sliders and numeric fields.

Key Fix: Integrated label mapping to ensure the app displays actual crop names (e.g., "RICE") instead of numerical indices.



DEVELOPMENT LIFECYCLE



Stage 1

EDA & Data
Preprocessing



Stage 2

Algorithm
Benchmarking



Stage 3

Streamlit App
Engineering



Stage 4

Deployment &
Tunneling



Questions & Discussion

Enabling sustainable farming through precision data analytics.

GitHub Repository:

github.com/BaintyKaur/Crop-Recommendation-System

Ashliya M Riyaz | Nivedh P D | Bainty Kaur Chugh

IMAGE SOURCES



Thumbnail for
[www.freepik.com](https://img.freepik.com)

https://img.freepik.com/premium-photo/diverse-team-scientists-collaborating-soil-plant-research-showcasing-power-innovation-agriculture_856795-81474.jpg

Source: www.freepik.com



<https://image.made-in-china.com/2f0j00eMfbRPVJhhkQ/Soil-Nutrient-Detector-Laboratory-Special-Instrument-Soil-Nitrogen-Phosphorus-and-Potassium-Analysis-Equipment.webp>

Source: m.made-in-china.com



Thumbnail
for

https://www.mdpi.com/mathematics/mathematics-10-04460/article_deploy/html/images/mathematics-10-04460-g004-550.jpg

Source: www.mdpi.com

www.mdpi.com



https://strapi.a-bots.com/uploads/smartphone_apps_in_agriculture_for_farm_advice_76f44d2b2f.jpg

Source: a-bots.com